

Midterm. Row 1. Solutions

Mathematical Analysis
1st Year

Babes-Bolyai University
Bachelor in Computer Science

Midterm Test

1. Prove that if the series $\sum_{n=1}^{\infty} x_n$ is convergent, then $\lim_{n \rightarrow \infty} x_n = 0$.

1p

Let $S_n = x_1 + x_2 + \dots + x_n$. $\sum_{n=1}^{\infty} x_n$ converges $\Leftrightarrow (S_n)$ convergent.

0.5p

If (S_n) convergent $\rightarrow \lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} (S_n - S_{n-1}) = 0$.

0.5p

2. Find the value of $\lim_{n \rightarrow \infty} \left(\frac{1}{3n+1} + \frac{1}{3n+2} + \dots + \frac{1}{4n} \right)$

1p

$$\bullet \sum_{k=1}^n \frac{1}{3+nk} = \frac{1}{n} \sum_{k=1}^n \frac{1}{3+\frac{k}{n}} \xrightarrow{0.5p} \int_0^1 \frac{1}{3+x} dx = \ln(3+x) \Big|_0^1 = \ln \frac{4}{3}. \quad 0.5p$$

• Or we can use the sequence $C_n = 1 + \frac{1}{2} + \dots + \frac{1}{n} - \ln n \rightarrow \gamma$.

$$\sum_{k=1}^n \frac{1}{3+nk} = C_{4n} - C_n + \ln 4n - \ln 3n = \underbrace{C_{4n} - C_n}_{\rightarrow 0} + \ln \frac{4}{3} \rightarrow \ln \frac{4}{3}.$$

3. Study the convergence of the following:

3p

(a) $\sum_{n=2}^{\infty} \frac{1}{n \ln n}$.

(b) $\int_0^{\pi} \frac{1}{\sin^2 x} dx$

(c) $\sum_{n=1}^{\infty} \frac{1 \cdot 4 \cdot \dots \cdot (3n-2)}{3^n n!}$.

(a) Using the integral test, the series has the same nature as 0.5p

$$\int_2^{\infty} \frac{1}{x \ln x} dx \stackrel{\substack{\uparrow \\ u = \ln x, du = \frac{1}{x} dx}}{=} \int_{\ln 2}^{\infty} \frac{1}{u} du = \ln u \Big|_{\ln 2}^{\infty} = \infty, \text{ hence it is divergent. } \quad 0.5p$$

• Or using the Cauchy condensation test: $\sum_{n=2}^{\infty} \frac{1}{n \ln n}$ has the same nature as $\sum_{n=2}^{\infty} 2^n \frac{1}{2^n \ln 2^n} = \sum_{n=2}^{\infty} \frac{1}{\ln 2^n}$ divergent.

(b) Improper integral at 0 and π where $\sin^2 x = 0$.

$$\int_0^{\pi} \frac{1}{\sin^2 x} dx = \int_{0+0}^{\frac{\pi}{2}} \frac{1}{\sin^2 x} dx + \int_{\frac{\pi}{2}}^{\pi-0} \frac{1}{\sin^2 x} dx.$$

Recall that $\int_{a+0}^b \frac{1}{(x-a)^p} dx, \int_a^{b+0} \frac{1}{(b-x)^p} dx \begin{cases} \text{converges if } p < 1 \\ \text{diverges if } p \geq 1 \end{cases} \quad 0.5p$

Since $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1, \lim_{x \rightarrow 0} \frac{\frac{1}{\sin^2 x}}{\frac{1}{x^2}} = 1$, hence $\int_{0+0}^{\frac{\pi}{2}} \frac{1}{\sin^2 x} dx$ "like" $\int_{0+0}^{\frac{\pi}{2}} \frac{1}{x^2} dx$, divergent. 0.5p

Also, $\lim_{x \rightarrow \pi} \frac{\sin x}{\pi-x} = \lim_{x \rightarrow \pi} \frac{\sin(\pi-x)}{\pi-x} = 1$, hence $\int_{\frac{\pi}{2}}^{\pi-0} \frac{1}{\sin^2 x} dx$ "like" $\int_{\frac{\pi}{2}}^{\pi-0} \frac{1}{(\pi-x)^2} dx$, divergent. 0.5p

Conclusion: $\int_0^{\pi} \frac{1}{\sin^2 x} dx$ is divergent.

$$(c) x_n = \frac{1 \cdot 4 \cdot \dots \cdot (3n-2)}{3^n \cdot n!}, \quad \frac{x_{n+1}}{x_n} = \frac{1 \cdot 4 \cdot \dots \cdot (3n-2) \cdot (3n+1)}{3^{n+1} (n+1)!} \cdot \frac{3^n n!}{1 \cdot 4 \cdot \dots \cdot (3n-2)} = \frac{3n+1}{3(n+1)} = \frac{3n+1}{3n+3} \rightarrow 1. \quad 0.5p$$

0.25 Raabe-Duhamel test, $n \left(\frac{x_n}{x_{n+1}} - 1 \right) = n \left(\frac{3n+3}{3n+1} - 1 \right) = n \cdot \frac{2}{3n+1} \rightarrow \frac{2}{3} < 1 \Rightarrow \text{divergent}$ 0.25p

4. Find the Taylor series around 0 and its radius of convergence for each of the following:

(a) $\ln(1-x)$.

1p (b) $\sqrt{1+x}$.

1p

(a) $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n \rightarrow \text{Taylor series around 0.}$ 0.25p

$$f(x) = \ln(1-x), \quad f(0) = 0, \quad f'(x) = \frac{1}{x-1}, \quad f'(0) = -1; \quad f''(x) = \frac{-1}{(x-1)^2}, \quad f''(0) = -1.$$

$$f^{(3)}(x) = \frac{2}{(x-1)^3}, \quad f^{(3)}(0) = -2; \quad f^{(4)}(x) = \frac{-2 \cdot 3}{(x-1)^4}, \quad f^{(4)}(0) = -3!.$$

$$f^{(n)}(x) = (-1)^{n+1} \frac{(n-1)!}{(x-1)^n}, \quad f^{(n)}(0) = -(n-1)!, \quad n \geq 1. \quad 0.5p$$

$$\text{Hence } \ln(1-x) = f(x) = f(0) + \sum_{n=1}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = \sum_{n=1}^{\infty} \frac{-(n-1)!}{n!} x^n = \sum_{n=1}^{\infty} -\frac{x^n}{n}.$$

• Radius of convergence: $R = \frac{1}{L}$, where $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{n}{n+1} = 1 \Rightarrow R = 1.$ 0.25p

• Or we can consider $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ and integrate: $-\ln(1-x) = \sum_{n=0}^{\infty} \frac{x^{n+1}}{n+1} = \sum_{n=1}^{\infty} \frac{x^n}{n}.$

(b) $f(x) = \sqrt{1+x} = (1+x)^{\frac{1}{2}}, \quad f(0) = 1; \quad f'(x) = \frac{1}{2}(1+x)^{-\frac{1}{2}}, \quad f'(0) = \frac{1}{2}.$

$$f''(x) = \frac{1}{2} \cdot \left(-\frac{1}{2}\right) \cdot (1+x)^{-\frac{3}{2}}, \quad f''(0) = \frac{1}{2} \cdot \left(-\frac{1}{2}\right).$$

$$f^{(n)}(x) = \frac{1}{2} \left(\frac{1}{2}-1\right)\left(\frac{1}{2}-2\right) \dots \left(\frac{1}{2}-n+1\right) (1+x)^{\frac{1}{2}-n}, \quad f^{(n)}(0) = \frac{1}{2} \left(\frac{1}{2}-1\right) \dots \left(\frac{1}{2}-n+1\right), n \geq 2$$

$$= (-1)^{n+1} \frac{1 \cdot 3 \cdot \dots \cdot (2n-3)}{2^n} \quad 0.5p$$

$$\sqrt{1+x} = f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = 1 + \frac{x}{2} + \sum_{n=2}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = 1 + \frac{x}{2} + \sum_{n=2}^{\infty} (-1)^{n+1} \frac{1 \cdot 3 \cdot \dots \cdot (2n-3)}{2^n \cdot n!} x^n.$$

0.25p

• Radius of convergence: $R = \frac{1}{L}$, where $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{2n-1}{2(n+1)} = 1 \Rightarrow R = 1.$ 0.25p

5. Consider the power series $\sum_{n=2}^{\infty} \frac{x^n}{n^2 - n}$.

(a) Find its radius of convergence and its convergence set.

1p

(b) Find its sum for any x in the convergence set.

1p

(a) $\sum_{n=2}^{\infty} \frac{1}{\underbrace{n(n-1)}_{a_n}} x^n$. Radius of convergence $R = \frac{1}{L}$, where $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = 1 \Rightarrow R = 1$. 0.5p

• for $x=1$: $\sum_{n=2}^{\infty} \frac{1}{n^2 - n}$ has the same nature as $\sum_{n=2}^{\infty} \frac{1}{n^2}$, which is convergent. 0.25p

• for $x=-1$: $\sum_{n=2}^{\infty} \frac{(-1)^n}{n^2 - n}$ is convergent by the Leibniz test since $\frac{1}{n^2 - n} \rightarrow 0$. 0.25p

(b) $\sum_{n=2}^{\infty} \frac{x^n}{n(n-1)} = \sum_{n=2}^{\infty} \left(\frac{1}{n-1} - \frac{1}{n} \right) x^n = \sum_{n=2}^{\infty} \frac{x^n}{n-1} - \sum_{n=2}^{\infty} \frac{x^n}{n} = x \cdot \sum_{n=2}^{\infty} \frac{x^{n-1}}{n-1} - \sum_{n=2}^{\infty} \frac{x^n}{n}$. 0.25p

• $\left(\sum_{n=2}^{\infty} \frac{x^n}{n} \right)' = \sum_{n=2}^{\infty} \left(\frac{x^n}{n} \right)' = \sum_{n=2}^{\infty} x^{n-1} = \sum_{n=1}^{\infty} x^n = x + x^2 + \dots = \frac{1}{1-x} - 1, \forall x \in (-1, 1)$. 0.25p

Integrating we obtain that $\sum_{n=2}^{\infty} \frac{x^n}{n} = -\ln(1-x) - x, \forall x \in (-1, 1)$. 0.25p

• $\sum_{n=2}^{\infty} \frac{x^{n-1}}{n-1} = \sum_{n=1}^{\infty} \frac{x^n}{n} = x + \sum_{n=2}^{\infty} \frac{x^n}{n} = x - \ln(1-x) - x = -\ln(1-x), \forall x \in (-1, 1)$.

• We get that $\sum_{n=2}^{\infty} \frac{x^n}{n(n-1)} = -x \ln(1-x) + \ln(1-x) + x =: g(x), \forall x \in (-1, 1)$. 0.25p

• Notice that g can also be defined for $x = \pm 1$, where the series converges, hence we have that $\sum_{n=2}^{\infty} \frac{1}{n(n-1)} = g(1) = \lim_{x \uparrow 1} \left[\underbrace{(1-x) \ln(1-x)}_{\rightarrow 0} + x \right] = 1$.

$\sum_{n=2}^{\infty} \frac{(-1)^n}{n(n-1)} = g(-1) = 2 \ln 2 - 1$. 0.25p